

AE6030 – Intro to data handling & computing



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Resources

2

- Understanding Space (Sellers)
 - Section 13.1, p. 449-460
- Satellite Platform Design (Berlin)
 - Chapter 8 – 8 pages only. *On your StudySpace.*
- Space Mission Analysis & Design (Wertz / Larsen).
 - 3rd edition: Chapter 16 – spacecraft computer systems.
 - P. 645-684

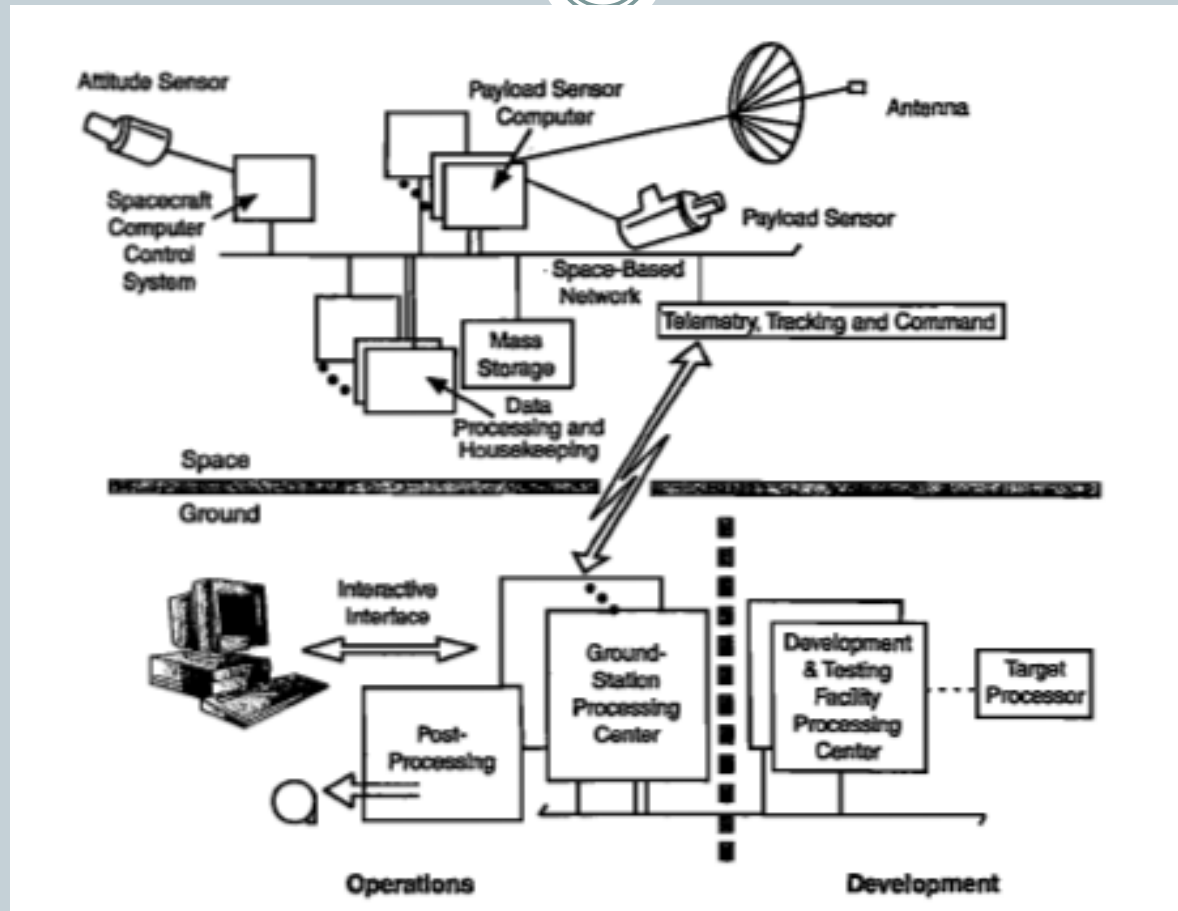
Learning objectives

3

- Data handling roles
- Unique requirements of Spacecraft computers
- Functions of an OBC
- Link between data handling and TTC
- What is TTC and why is it vital: uplink and downlink
- Sensor & transducer requirements
- Impact of radiation: need for autonomy
- OBDH design checklist

Mission computer systems

4



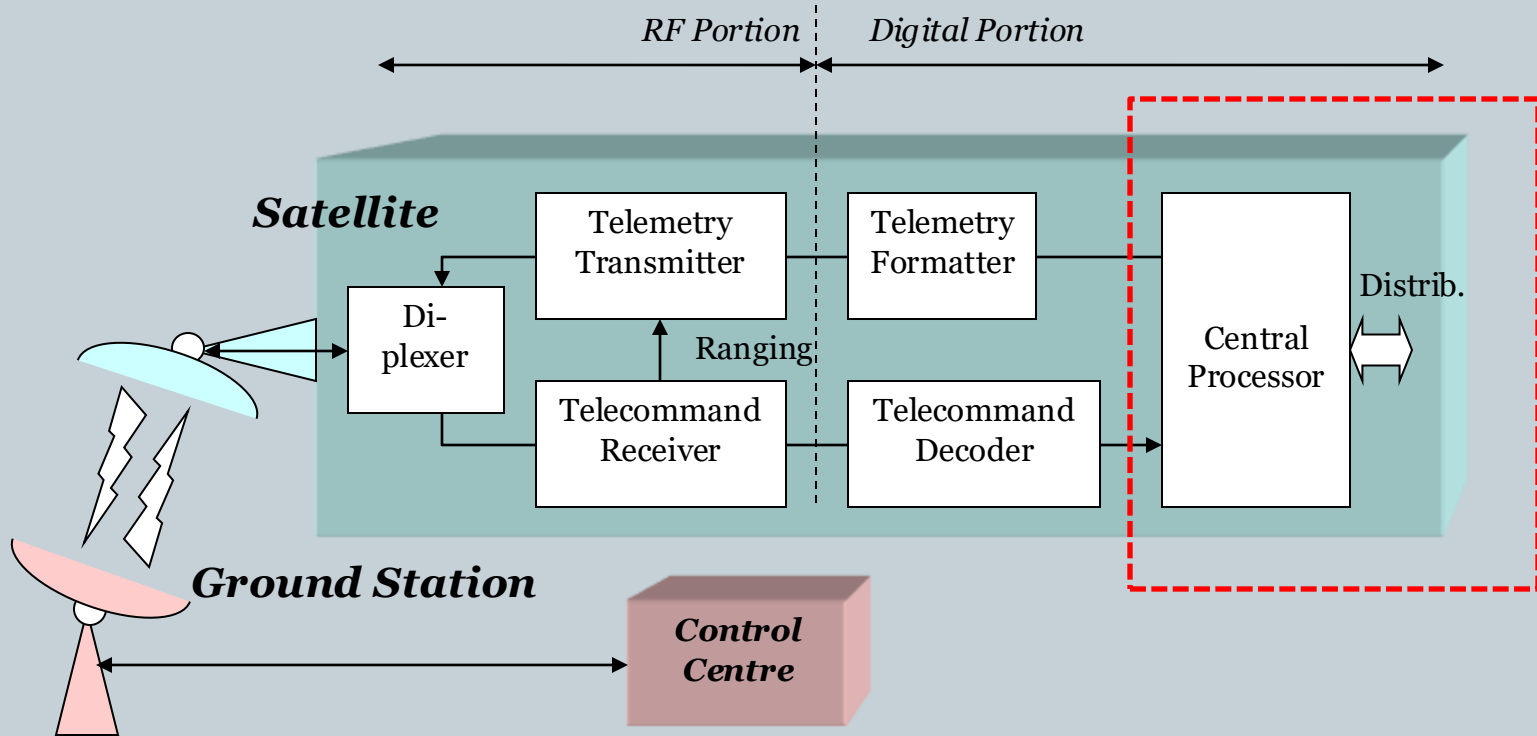
Spacecraft data handling functions



- Incoming **telecommands**: Checking legitimacy and routing to the intended subsystem for execution.
- Outgoing **telemetry**: Collection of telemetry parameters from the subsystems and packaging for onward transmission.
- **Storage** of telecommands for later execution.
- Storage of telemetry for later transmission.
- Implementation of **emergency** actions (e.g. switching from a defective unit to a redundant unit).
- Generation of a **clock** signal for onboard activity synchronisation.
- Autonomous attitude & orbit **determination**.
- **Autonomous execution** of operational scripts uplinked by the control centre.

Architecture: link to TTC

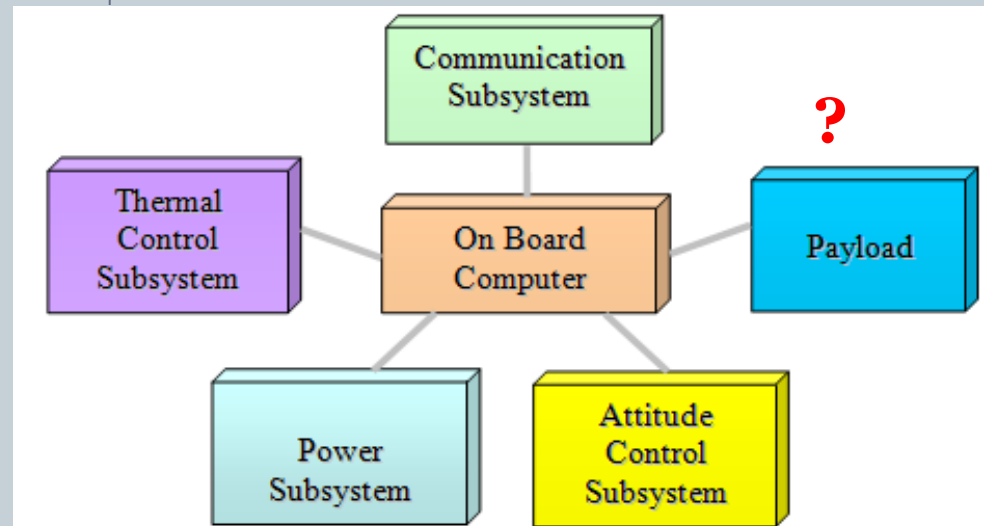
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Satellite Subsystems

7

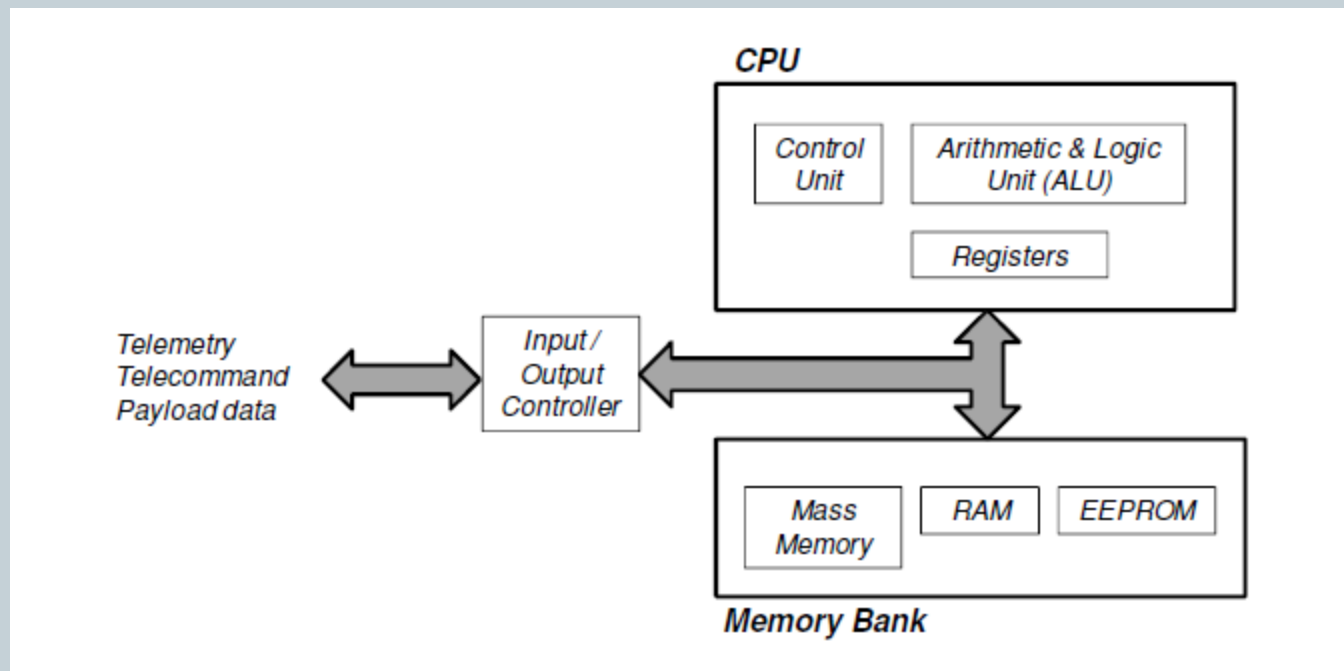
- OBDH
 - OBC
 - TT&C
- Electrical Power system
 - Energy sources
 - Energy storage
 - Power distribution
 - Power regulation & Control
- Attitude and orbit control
- Thermal
- Structure



One missing subsystem?

Simplified OBDH architecture

8

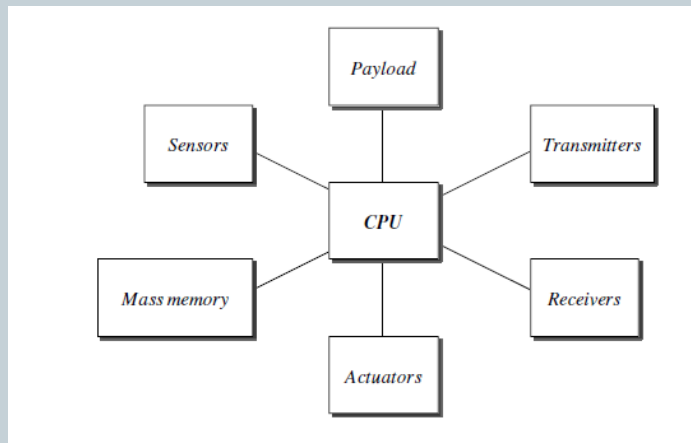


Berlin – SPLAT v. 6 (2013)

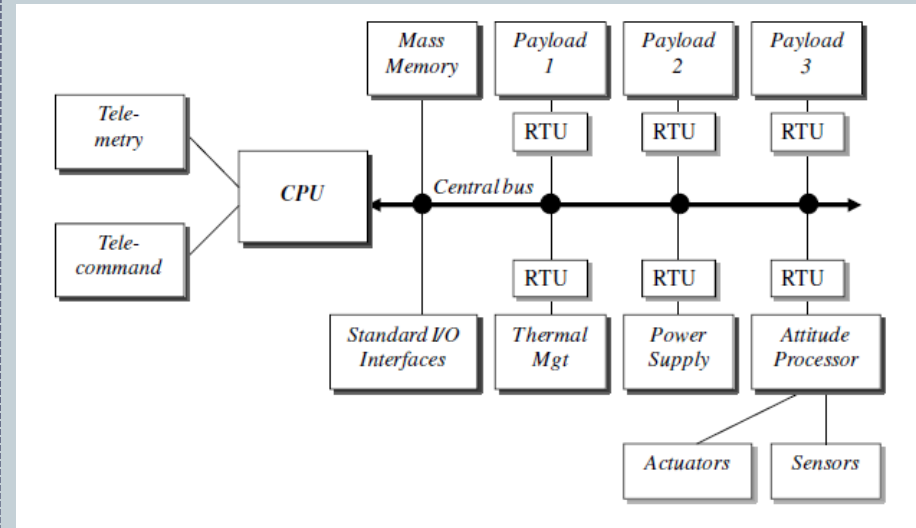
Architecture options

9

- Centralised

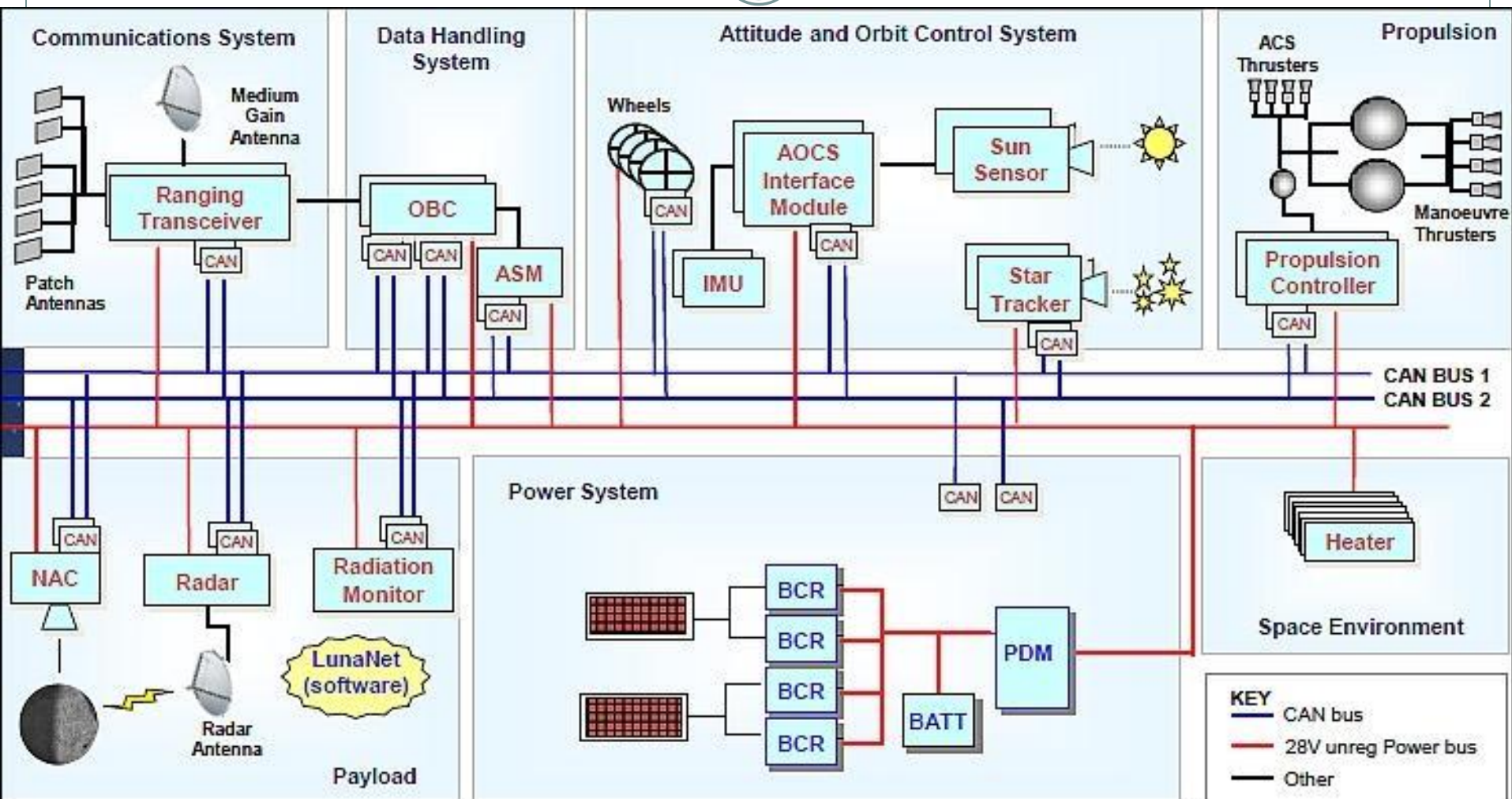


- Distributed



Satellite Subsystems

10



OBDH: to remember

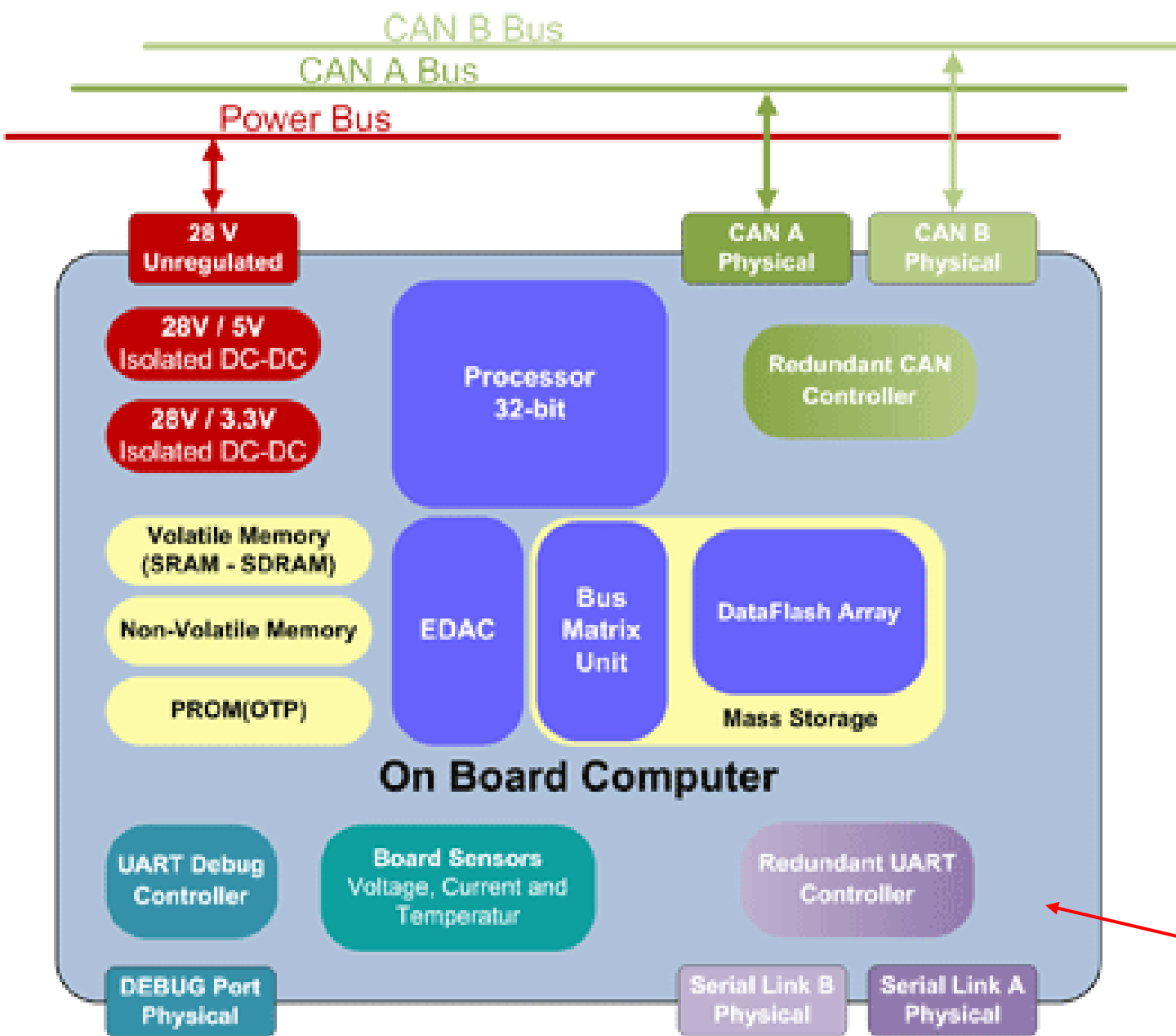
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- OBDH
 - Subsystem specific applications & generic functions
 - One or more processing unit(s)
 - TM & TC services and interfaces with the RF communication chain
 - Time synchronisation and distribution
 - Built-in fault tolerance
- Communicates with subsystems and payloads via a 'bus' e.g. MIL-1553B standard, CAN, Spacewire,

On-Board Computers

13

PROCESSING DATA ON-BOARD



**Compare
to
typical PC!**



Payload

On-Board Computers

15

- Two or more
 - Primary and secondary
 - Sus-systems and payload may have one

Redundancy

16

- What is meant by redundancy?
 - Is ...
- How can it be implemented?
 - Think in terms of components

Backup with different approaches / backup with ground systems / **Duplicate & cross strapping.**

- 2 identical processors. Hot or cold redundancy / watchdogs.
- 3 processors voting scheme

OBC software

17

- **Components**
 - Operating system software
 - Control system tasks
 - System management tasks
 - Mission data software
- **Critical software - ?**

OBDH

18

TELEMETRY AND TELECOMMAND

Telemetry, Tracking & Control (TT&C)

19

- Think of a remote controlled car!
 - How does it work?
 - What are the main components?
 - What is missing!
- TT&C subsystem communicates to ground spacecraft vital signs
- TT&C controls spacecraft functions and condition from ground station
 - reconfigures satellite or subsystems

Telemetry, Tracking & Control (TT&C)

20

- **Function**

- TC: Receives, validates, decodes, distributes commands
- TC: collects, processes, formats data ready for downlinking
- Other housekeeping duties: timekeeping, health monitoring, ...

- **Design**

- What needs monitoring? Determines number of TLM channels
- What needs controlling? And how (delayed, real time)?
Determines number of cmds, types, ..

Telemetry, Tracking & Command (T&C)

21

Telemetry

- Measure and relay internal measurements e.g.
 - voltages, currents, temperatures, accelerations, valve and relay states
- External measurements e.g. B-field, etc..
- Measured value could be analogue or digital
 - **What is implication?**
- Could be 100s of values
 - ?

Telecommand

- Control satellite
 - Low data rate, store, verify, execute or execute on time, programmable control
- Could be a single pulse or a word (several bits)
- Command must be decoded
 - ? how
- Command may be verified

Telemetry, Tracking & Control (TT&C)

22

- Command timing – when to send / execute a command
- Examples:
 - Deep space mission – round trip delay is 20 minutes, think of example when you would issue one of the 3 command types:
 - ✦ Immediate
 - ✦ Delayed
 - ✦ Priority driven

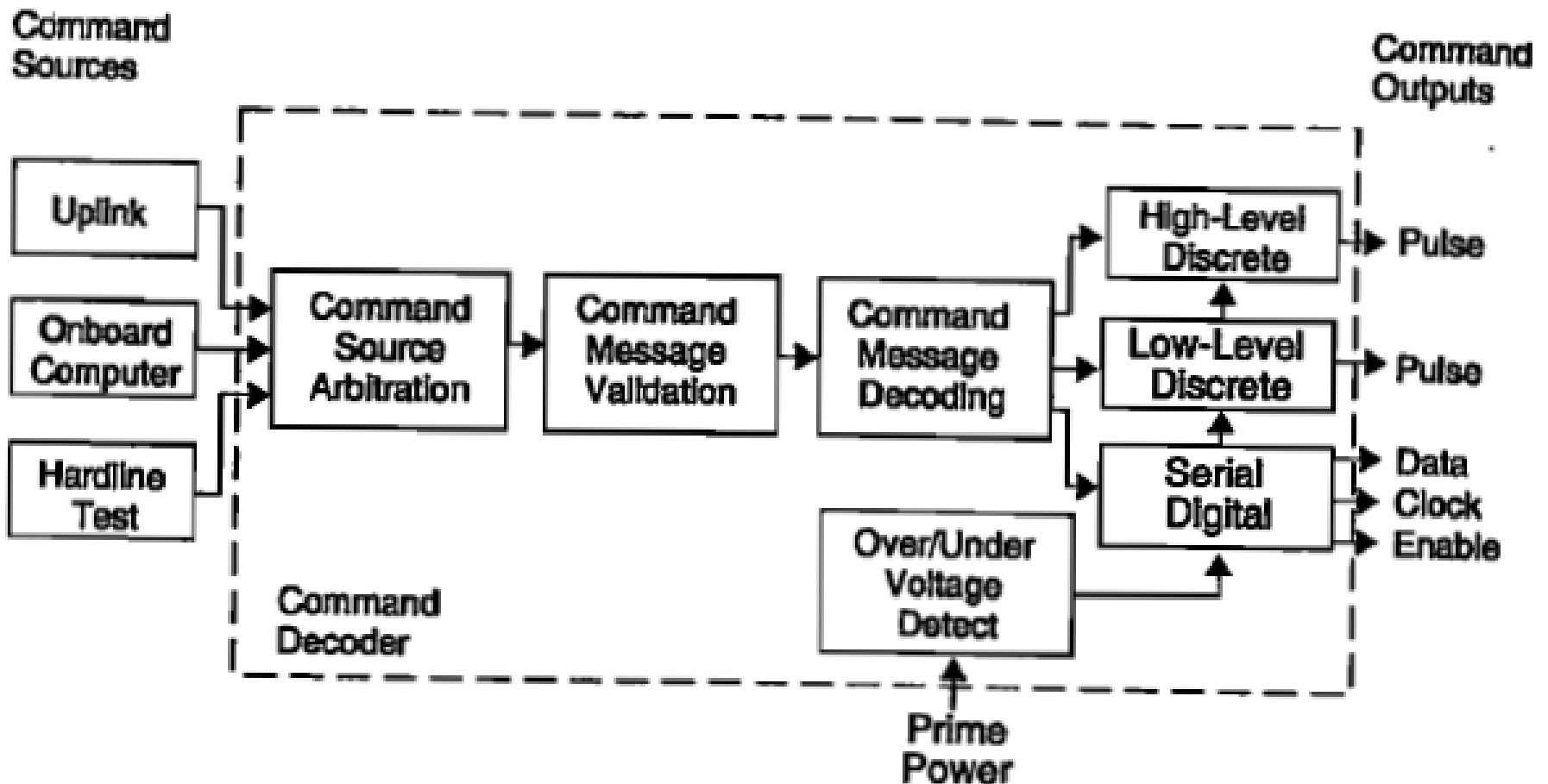
TTC (OBDR) summary

23

- Receive commands from Command and Data Handling (CD&H) subsystem
- Provide health and status data to CD&H
- Perform antenna pointing
- Perform mission sequence operations
- Autonomously select omni-antenna when spacecraft attitude is lost
- Autonomously detect faults and recover communications using stored software sequence

Data Handling: uplink

24



Data Handling: uplink

25

- Arbitration
 - Select input
- Validation
 - Correct address
 - EDAC
 - Valid command
 - Valid timing
 - Authenticated
- Decoding

OBDH

26

- Message format (CCSDS)
 - Synch bits, s/c address, cmd, crc

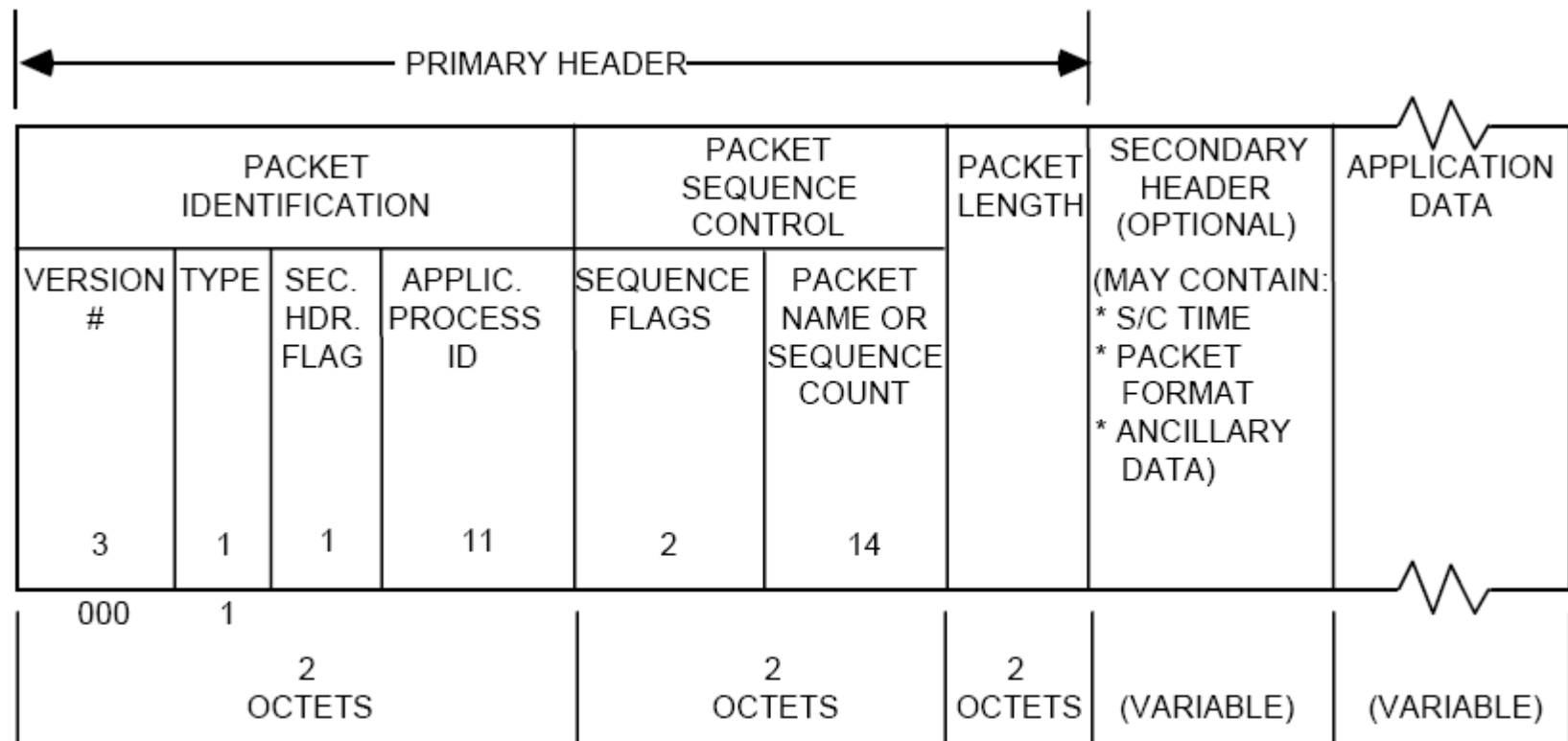
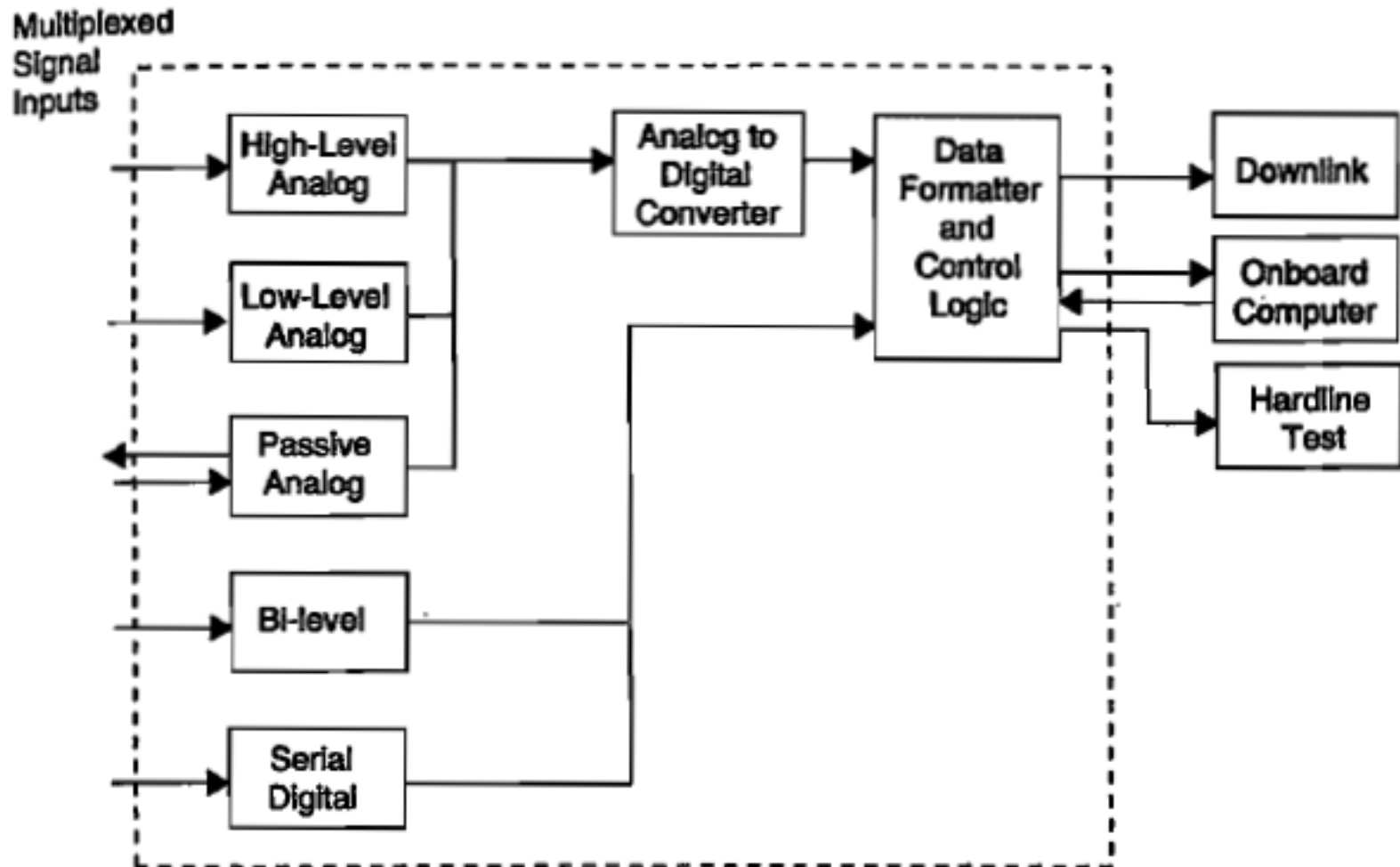


Figure 5-1: Telecommand Packet Format

Data Handling: downlink

28



Data Handling: downlink

29

- **Telemetry**
 - Readings from sensors (internal and external observations)
 - Amplification and convert to digital if required
 - Time division multiplexing
 - Formatted data sent to downlink or OBC

Telemetry Processing

30

- Compression
- Analysis for autonomous systems
- Formatting
- Storage

Sensors and Transducers

31

TT&C

Sensors and Transducers

32

- Sensors change state as a function of an external event
- Transducers convert energy from one form to another
- Outputs can be
 - Resistance
 - Capacitance
 - Current
 - Voltage

Analogue to Digital conversion

33

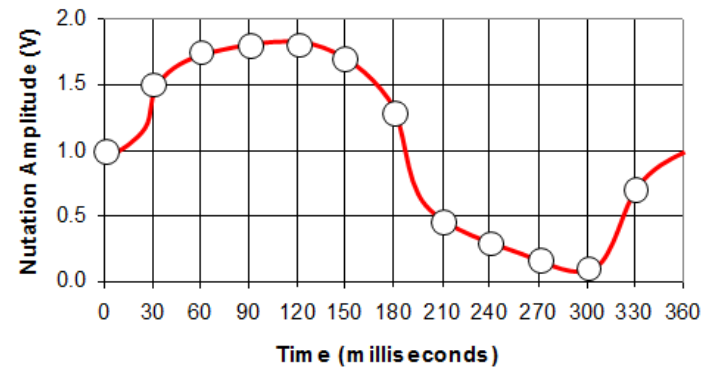
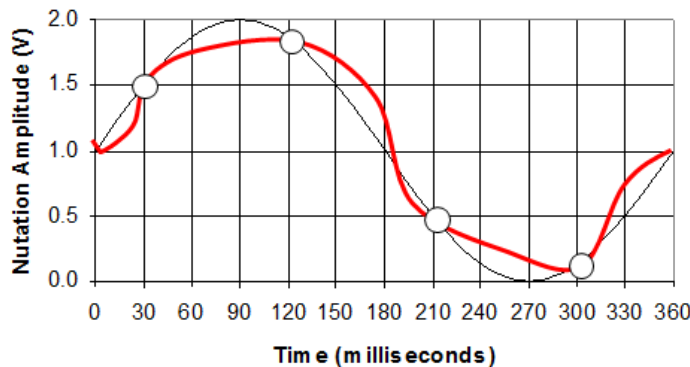
Type	Conversion Rate	Word Size	Power
High Speed ADC	$50 \times 10^6 / \text{sec}$	8 bit	2.5 W
High Resolution ADC	$1 \times 10^5 / \text{sec}$	16 bit	1.5 W
Low Power ADC	$2.5 \times 10^4 / \text{sec}$	8 bit	0.005 W

Converts a voltage into a discrete value between $2^n - 1$
Quantisation error decreases with number of bits, n .

Signal Conditioning and Selection

35

- Conditioning ensures proper level, dynamic range, frequency response, impedance, ground reference, common mode rejection
 - Commutation selects the proper sensor at a given time
- Sampling frequency determined by Nyquist criteria



Telemetry, Tracking & Control (TT&C)

37

- **Summary**

- two-way communication link between the satellite and ground control
- ground station tracks satellite's position
- controls satellite subsystems - propulsion, thermal, etc
- monitors the temperature, electrical voltages, and other important parameters of a satellite.

OBDH Design

38

- **What if command decoding fails? How can we mitigate?**
 - Interfaces to other subsystems must protect the command decoder
 - No commands or transient signals may appear on command outputs during application or removal of prime power or during under/over voltage conditions
 - If a commands integrity is in doubt, reject it
 - Multiple commands are required for critical/ dangerous operations
 - No single component failure can result in unintended operation
 - No commands shall interrupt the uplink source to the command decoder

OBDH for remote sensing

39

Remote sensing satellite

Design exercise – think about components to include
- consider each subsystem separately....

TT&C Interfaces

40

Subsystem	Requirement
Attitude Determination and Control	Antenna pointing
Command and Data Handling	Command and telemetry data rates Clock, bit sync, and timing requirements Two-way comm requirements Autonomous fault detection and recovery Command and telemetry electrical interface
Electrical Power Subsystem	Distribution requirements
Thermal/Structural	Heat sinks for TWTAs Heat dissipation of all active boxes Location of TT&C subsystem electronics Clear field of view and movement for all antennas
Payload	Storing mission data RF and EMC interface requirements Special requirements for modulation and coding

October 29, 2003

Massachusetts Institute of
Technology

9

Radiation Effects

41

DELETERIOUS EFFECTS OF RADIATION ON SPACECRAFT SYSTEMS

Radiation effects on bus systems

42

- Gamma rays leave a trail of charged particles in their wake
 - Measured in Rad(Si)
- High energy neutrons cause structural damage in solid state materials
- High energy charged particles generate clouds of electrical charges
- Charge on gates of metal-oxide semiconductors can change its state – single event upset
- CMOS can latch up across the power supply

Error detection and protection



- Error Detection
 - recognise that an error has happened
- Error Correction
 - repair damaged data
- parity and CRC
- BCC and Hamming

Error detection

44

- Parity checking
 - Add an extra bit to the word so that the number of logic ones is always even
 - When a word is read out of memory the logic ones are counted
- Checksum for words
 - Logic ones are summed modulo 2^n to form n-bit memory checksum
 - Checksum is calculated before transmission and sent with the data
 - Checksum is calculated upon receipt

Error Detection and Correction

45

- Hamming codes put parity bits at special locations
 - Each data bit is checked by a unique set of parity bits
 - Permits error detection and correction
 - Requires k parity bits for an n -bit data word

Fault tolerance

46

- **Redundancy**
 - Duplicate equipment
 - Backup with different approaches
 - Backup with ground systems
 - Data bus delivery systems
 - Inter-changeable components

Fault tolerance

47

- Distributed processing
 - Allocates software processing to any one of several processors depending on
- Mission phase
- Hardware availability
- Subsystem failures

Fault tolerance

48

- Multiple execution
 - Master/slave CPUs
 - Separate algorithms
 - Certification trails
- Fault roll-back – return to a saved context saved before the fault occurred
- Watchdog timers
- Improper sequence detectors

Spacecraft autonomy

49

- What is autonomy?
 - "ability of integrated sensing, perceiving, analysing, communicating, planning, decision-making, and acting/executing, to achieve its goals as assigned"
- Why autonomy?
 - Complex system, complex environment
 - **Benefits?**
- Levels of autonomy
 - 6 to 8 levels
 - NIST developing standards

Spacecraft autonomy

50

1. Human operated
2. Human assisted
3. Human delegated
4. Human supervised
5. Mixed initiative
6. Fully autonomous

Levels of autonomy

51

	Name	Description
1	Human operated	All activity within the system is the direct result of human-initiated control inputs. The system has no autonomous control of its environment, although it may have information-only responses to sensed data.
2	Human assisted	The system can perform activity in parallel with human input, acting to augment the ability of the human to perform the desired activity, but has no ability to act without accompanying human input. An example is automobile automatic transmission and anti-skid brakes.
3	Human delegated	The system can perform limited control activity on a delegated basis. This level encompasses automatic flight controls, engine controls, and other low-level automation that must be activated or deactivated by a human input and act in mutual exclusion with human operation.
4	Human supervised	The system can perform a wide variety of activities given top-level permissions or direction by a human. The system provides sufficient insight into its internal operations and behaviours that it can be understood by its human supervisor and appropriately redirected. The system does not have the capability to self-initiate behaviours that are not within the scope of its current directed tasks.
5	Mixed initiative	Both the human and the system can initiate behaviours based on sensed data. The system can coordinate its behaviour with the human's behaviours both explicitly and implicitly. The human can understand the behaviours of the system in the same way that he understands his own behaviours. A variety of means are provided to regulate the authority of the system with respect to human operators.
6	Fully autonomous	The system requires no human intervention to perform any of its designed activities across all planned ranges of environmental conditions.

OBDH design aspects

52

- What tasks are to be performed?
- What is nature and rate of data flow?
- Architecture options?
- Tasks: real time v. deferred time? Hardware / software / firmware implementation?
- Operating system & programming language options?
- Processing & memory capacity: Processor speeds
 - Margins?
- Environmental protection, fault tolerance
- Volume, mass, power, reliability?
- Testing?
- Availability of units – cost - delivery

Recap on learning objectives

53

- Data handling role among other sat. subsystems
- Unique requirements of Spacecraft computers
- Functions of an OBC
- Link between data handling and TTC
- What is TTC and why is it vital: uplink and downlink
- Sensor & transducer requirements
- Impact of radiation: need for autonomy
- OBDH design checklist